

Measurement-Based Modeling of V2V Fading Channels

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Academic Year 2025–2026

The MATLAB source code used for the channel modeling, simulation, and performance evaluation is available

HERE

1 Introduction

Vehicle-to-vehicle (V2V) communication has emerged as a key enabling technology for intelligent transportation systems, with applications ranging from road safety and collision avoidance to traffic efficiency and autonomous driving. In V2V systems, vehicles communicate directly with each other without relying on fixed infrastructure, which makes the wireless propagation channel highly dynamic and strongly dependent on the surrounding environment.

One of the main challenges in V2V communication is the presence of multipath propagation combined with high mobility of both the transmitter and receiver. These effects result in fast time-varying fading, significant Doppler shifts, and rapid fluctuations of the received signal strength. Accurate modeling of the V2V fading channel is therefore essential for system design and performance evaluation.

In this work, a measurement-based approach is adopted to model V2V fading channels in urban and highway environments. Measured cumulative distribution functions (CDFs) of the fading process are used to fit classical Rayleigh and Rician channel models. The identified models are then simulated in the time domain using the sum-of-sinusoids method, and their impact on the performance of a BPSK communication system is evaluated.

2 Measurement Setup

2.1 Measurement Setup Overview

The vehicle-to-vehicle channel measurements were conducted using the setup summarized in Table 1. The carrier frequency was set to 5.2 GHz and both the transmitter and receiver were equipped with $\lambda/4$ monopole antennas.

2.2 Coaxial Cable Loss Estimation

The connection between the antenna and the transceiver is realized using a 2 m coaxial cable. At a carrier frequency of 5.2 GHz, typical RF coaxial cables exhibit an attenuation of approximately 0.5 dB/m. Therefore, the total cable loss can be estimated as

$$L \approx 2 \times 0.5 = 1 \text{ dB}.$$

2.3 Low Noise Amplifier Characteristics

In addition to its gain, an important characteristic of the Low Noise Amplifier (LNA) is its noise figure. The noise figure quantifies the amount of noise introduced by the amplifier relative to an ideal noiseless amplifier. Since the LNA is the first active component in the receiver chain, its noise figure has a major impact on the overall receiver sensitivity and signal-to-noise ratio. For this reason, LNAs are designed to provide a low noise figure while maintaining sufficient gain.

2.4 Maximum Doppler Frequency

The maximum Doppler frequency shift is given by

$$f_{D,\max} = \frac{v}{c} f_c,$$

where $f_c = 5.2 \text{ GHz}$ is the carrier frequency and $c \approx 3 \times 10^8 \text{ m/s}$ is the speed of light. Assuming typical vehicle speeds of $v_{\text{urban}} = 50 \text{ km/h}$ and $v_{\text{highway}} = 120 \text{ km/h}$, we obtain:

$$f_{D,\max}^{(\text{urban})} \approx 241 \text{ Hz}, \quad f_{D,\max}^{(\text{highway})} \approx 578 \text{ Hz}.$$

2.5 Coherence Time

An approximate coherence time can be related to the maximum Doppler frequency by

$$T_c \approx \frac{0.423}{f_{D,\max}}.$$

Using $f_{D,\max}^{(\text{urban})} \approx 241$ Hz and $f_{D,\max}^{(\text{highway})} \approx 578$ Hz (typical speeds assumed since the proposal does not specify them), we obtain

$$T_c^{(\text{urban})} \approx 1.76 \text{ ms}, \quad T_c^{(\text{highway})} \approx 0.73 \text{ ms}.$$

2.6 Vehicle Speed Assumptions

The project proposal does not explicitly specify the vehicle speeds in the urban and highway scenarios. Therefore, typical representative values are assumed for the purpose of Doppler and coherence-time calculations:

$$v_{\text{urban}} = 50 \text{ km/h} \approx 13.89 \text{ m/s}, \quad v_{\text{highway}} = 120 \text{ km/h} \approx 33.33 \text{ m/s}.$$

These values are then used to compute the maximum Doppler frequency and coherence time.

2.7 Number of Samples

The measurement duration is $T = 30 \text{ min} = 1800 \text{ s}$. The number of samples used to estimate the CDF is

$$N = f_s T.$$

For the urban scenario ($f_s = 1 \text{ kHz}$), this gives

$$N_{\text{urban}} = 1000 \times 1800 = 1.8 \times 10^6 \text{ samples}.$$

For the highway scenario ($f_s = 2 \text{ kHz}$), this gives

$$N_{\text{highway}} = 2000 \times 1800 = 3.6 \times 10^6 \text{ samples}.$$

3 Channel Modeling

3.1 Measured CDFs

Figures 1 and 2 show the measured cumulative distribution functions of the fading envelope for the urban and highway environments, respectively. The urban scenario exhibits a more severe fading behavior, while the highway scenario shows a steeper CDF, suggesting the presence of a dominant propagation component.

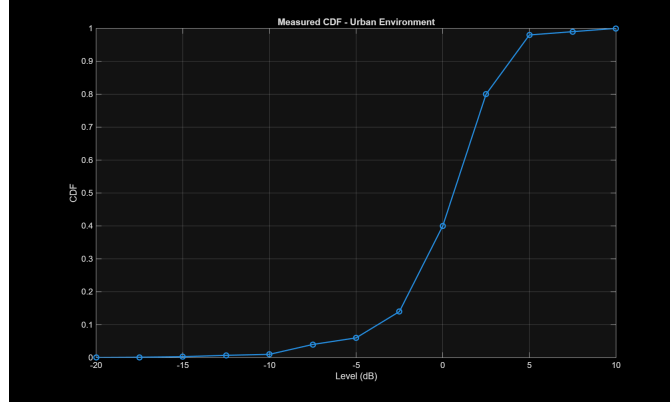


Figure 1: Measured CDF of the fading process in the urban environment.

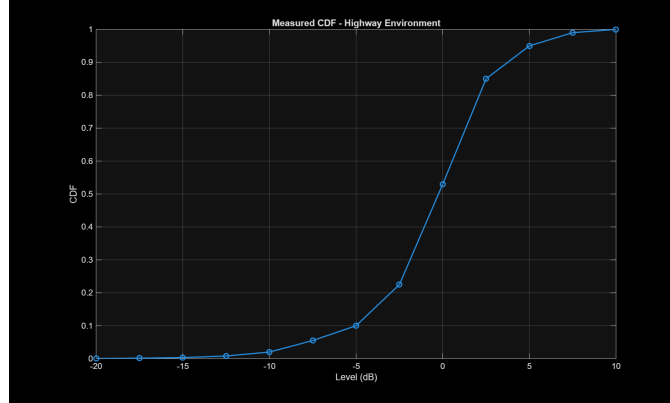


Figure 2: Measured CDF of the fading process in the highway environment.

3.2 Least-Squares Fitting of Fading Models

The measured CDFs were approximated using Rayleigh and Rician fading models. A least-squares criterion was employed to minimize the squared error between the measured and theoretical CDFs. The fitting was performed separately for the urban and highway scenarios.

3.3 Optimized Model Parameters

For the Rayleigh model, the optimized scale parameters were $\sigma_{\text{urban}} = 0.902$ and $\sigma_{\text{highway}} = 0.830$.

For the Rician model, the optimized parameters were:

$$\begin{aligned} \text{Urban: } s &= 1.025, \sigma = 0.323, K = 5.03 \text{ (7.02 dB)} \\ \text{Highway: } s &= 0.919, \sigma = 0.349, K = 3.48 \text{ (5.42 dB)} \end{aligned}$$

where $K = \frac{s^2}{2\sigma^2}$ is the Rician K -factor.

3.4 Rayleigh Model Fitting

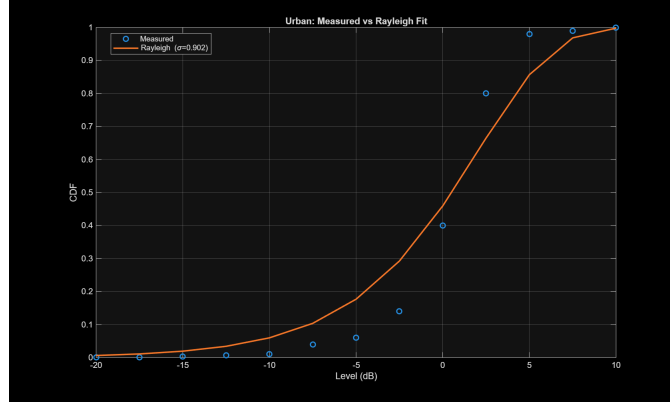


Figure 3: Measured CDF versus Rayleigh fit for the urban environment.

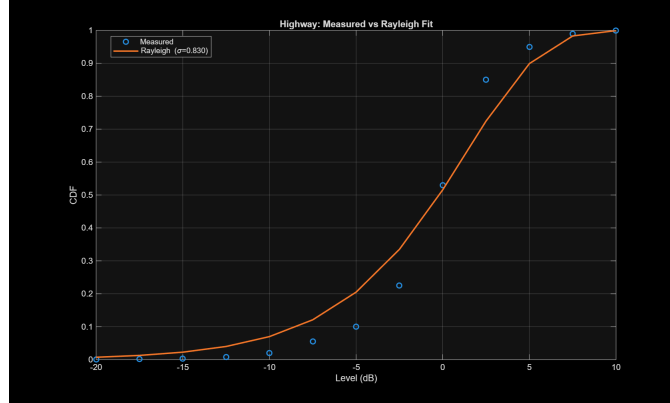


Figure 4: Measured CDF versus Rayleigh fit for the highway environment.

3.5 Measured and Theoretical CDF Comparison

Figures 5 and 6 compare the measured CDFs with the fitted Rayleigh and Rician models. The Rician model provides a noticeably better fit for the highway environment, particularly in the mid-amplitude region.

3.6 Suitable Fading Model

Based on the fitting results and physical interpretation of the propagation environment, Rayleigh fading is considered suitable for the urban scenario due to the absence of a dominant line-of-sight component. In contrast,

the highway scenario is better modeled by a Rician distribution, reflecting the presence of a strong line-of-sight path.

3.7 Rayleigh and Rician Model Comparison

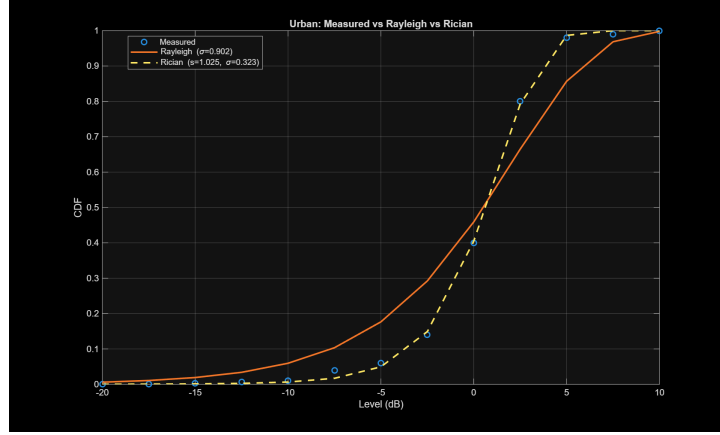


Figure 5: Measured CDF compared with Rayleigh and Rician models in the urban environment.

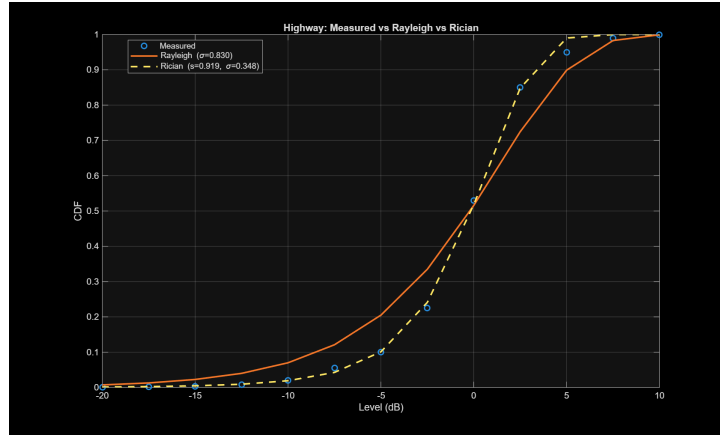


Figure 6: Measured CDF compared with Rayleigh and Rician models in the highway environment.

4 Simulation Methodology

The time-varying multipath fading channel was simulated using the sum-of-sinusoids method described in the Appendix. The in-phase and quadrature components of the channel were generated as sums of sinusoidal waves with Doppler-shifted frequencies and random phases.

For the urban environment, a Rayleigh fading model was assumed, while for the highway environment a Rician model was adopted by adding a deterministic line-of-sight component. The Doppler frequencies used in the simulation were derived from the estimated vehicle speeds. The resulting fading envelopes exhibit realistic temporal variations consistent with the statistical models identified in the previous section.

4.1 Simulation Methodology

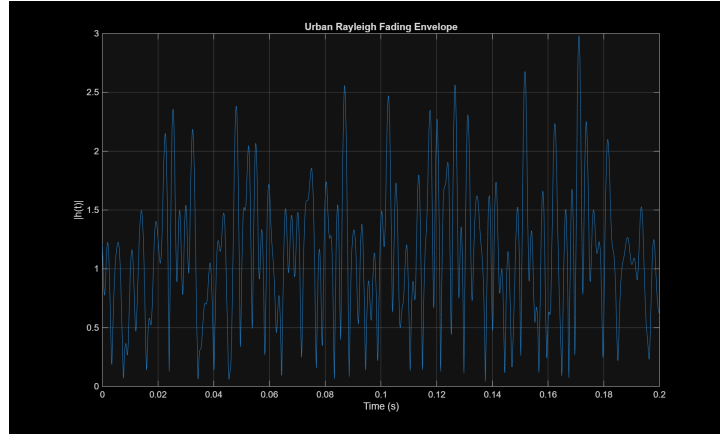


Figure 7: Time-domain Rayleigh fading envelope for the urban environment generated using the sum-of-sinusoids method.

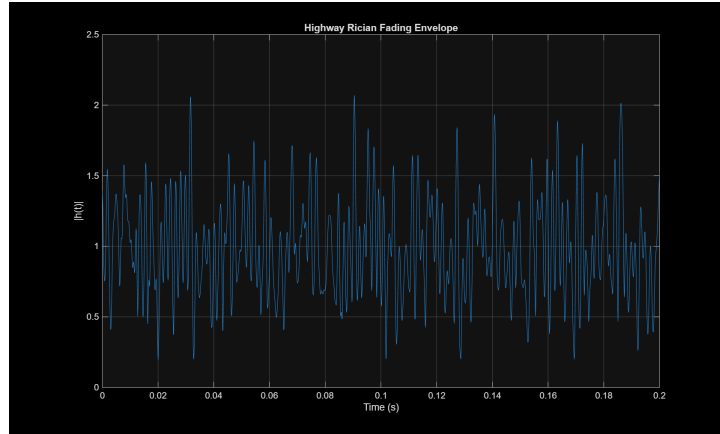


Figure 8: Time-domain Rician fading envelope for the highway environment generated using the sum-of-sinusoids method.

5 Results and Discussion

The performance of a BPSK communication system was evaluated over the simulated vehicle-to-vehicle fading channels. For both urban and highway environments, random binary sequences were modulated using BPSK, transmitted through the corresponding fading channel, and corrupted by additive white Gaussian noise. Coherent detection was assumed, with perfect channel state information at the receiver.

Figure 9 shows the resulting bit error rate (BER) as a function of E_b/N_0 . The urban scenario, modeled using a Rayleigh fading channel, exhibits the worst performance due to the frequent occurrence of deep fades. In contrast, the highway scenario, modeled using a Rician fading channel, achieves significantly better performance thanks to the presence of a dominant line-of-sight component. These results highlight the strong impact of multipath propagation characteristics on the reliability of V2V communications.

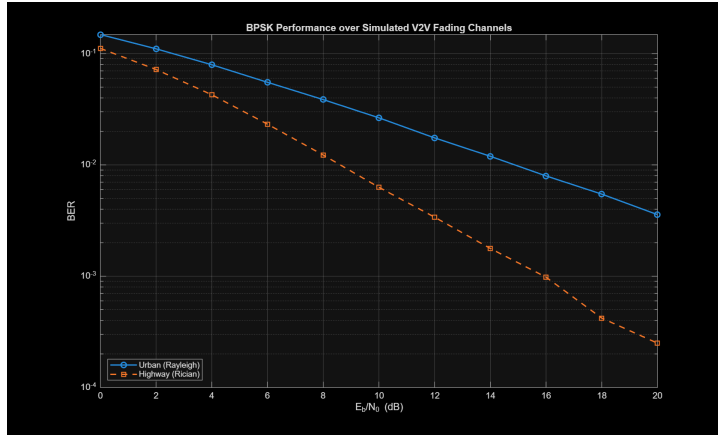


Figure 9: BER performance of BPSK over the simulated urban (Rayleigh) and highway (Rician) V2V channels.

6 Conclusion

In this work, a measurement-based modeling approach was used to characterize vehicle-to-vehicle fading channels in urban and highway environments. Measured CDFs were fitted using Rayleigh and Rician statistical models, leading to the identification of Rayleigh fading as suitable for urban scenarios and Rician fading for highway scenarios.

Time-domain channel realizations were generated using the sum of sinusoids method, incorporating Doppler effects due to vehicle mobility. The simulated channels were then used to evaluate the performance of a BPSK communication system. Simulation results demonstrated that the highway

environment provides superior performance compared to the urban environment, owing to the presence of a line-of-sight component and reduced fading severity. This study confirms the importance of accurate channel modeling for the design and performance evaluation of V2V communication systems.